Notes: Mullainathan, Schwartzstein, and Shleifer (QJE, 2008)
Coarse Thinking and Persuasion

Contents

1 Big picture

- **Question.** How can messages that contain no objective information still persuade consumers, voters, investors, or audiences? Why do slogans, brands, irrelevant product attributes, and selective presentation of facts change beliefs?
- **Core idea.** Individuals may think coarsely: instead of building a separate model of inference for every situation, they group situations into categories and apply the same inferential model to all situations inside a category.
- **Two mechanisms.** The paper separates **transference** from **framing**. Transference means that a message useful in one co-categorized situation is transferred to another situation where it is not objectively useful. Framing means that a message changes the category through which the audience interprets public information.
- **Persuader's problem.** A persuader observes the message that the audience would otherwise receive and may pay a cost to replace it with another message. For Bayesians, uninformative messages do not move beliefs. For coarse thinkers, uninformative messages may move beliefs by triggering transference or recategorization.
- **Key distinction from standard cheap talk.** The paper is not primarily about strategic provision of information. It is about strategic use of categorization mistakes. The persuader can influence the model the audience uses to interpret information.
- **Applications.** The theory explains why irrelevant product attributes can work, why branding can create value, why advertising can be persuasive without data, and why mutual fund ads report past returns in bull markets but withhold them after market declines.
- **Empirical illustration.** The paper uses financial advertisements in *Business Week* and *Money* and shows that the share of stock mutual fund ads reporting past returns is strongly procyclical with lagged market returns. It also shows that T. Rowe Price does not advertise good relative performance in down markets, consistent with a framing mechanism.
- **Economic takeaway.** Persuasion can operate by changing analogies and mental categories, not merely by providing information. Messages matter because they determine how existing information is interpreted.

2 Incremental contribution of the paper

- **Contribution relative to information-based persuasion models.** Standard economic models of persuasion focus on information revelation, costly signaling, disclosure, or cheap talk. This paper models persuasion that can be powerful even when the message is objectively uninformative in the target situation.
- **Formalization of associative thinking.** The paper brings ideas from psychology, marketing, linguistics, and political rhetoric into a formal economic model. It gives structure to metaphors, analogies, representativeness, and framing by representing them as category-based inference.
- **Two distinct channels of noninformative persuasion.** The paper separates transference and framing. This distinction is conceptually important because the first exploits an existing analogy, whereas the second creates or primes the analogy itself.
- **A tractable deviation from Bayesian inference.** The model nests Bayesian reasoning as the limiting case in which each situation is its own category. Coarse thinking is a disciplined modification: agents choose a category and then apply one inferential model to the whole category.
- **Strategic incentives for message creation, withholding, and branding.** The paper shows why persuaders may fabricate favorable attributes, omit favorable information, or create product brands. These behaviors arise from rational choice by the persuader interacting with non-Bayesian audience cognition.

- **Mutual fund advertising evidence.** The empirical section moves the theory from abstract examples to a real market in which persuasion is economically important. The key novelty is showing that funds with good relative returns may still avoid reporting them when reporting returns would prime a bad category.
- **Bridge between behavioral economics and IO/finance.** The framework helps explain advertising content, product differentiation, branding premia, and financial persuasion in markets with sophisticated persuaders and psychologically constrained audiences.

3 Model objects and measurement

3.1 Situations, quality, and public information

- The individual evaluates the quality of an object in situation $s = 0$, such as buying a shampoo, choosing a mutual fund, or evaluating a political candidate.
- The object has quality $q \in Q$, where quality can be interpreted as product quality, investment quality, candidate quality, or any payoff-relevant characteristic.
- The individual observes public information $r \in \{u, d\}$, where u and d can be read as favorable/unfavorable public information. In the mutual fund application, r corresponds to aggregate market performance.
- The individual also receives a message $m \in \{a, b\}$. In the persuasion situation, m is chosen by the persuader after observing private signal $x \in \{a, b\}$.

Interpretation: The setup separates information the persuader controls from public information the persuader cannot control. Persuasion works by changing how the audience interprets both.

3.2 Bayesian benchmark

A Bayesian evaluates the target situation $s = 0$ using the correct situation-specific model:

$$E[q \mid r, m, s = 0] = \sum_{x \in \{a,b\}} E[q \mid r, x, s = 0]\, \widehat{p}(x \mid r, m, s = 0).$$

The Bayesian reaction to message m is

$$E[q \mid r, m, s = 0] - E[q \mid r, s = 0] = (\widehat{p}(a \mid r, m, s = 0) - p(a \mid r, s = 0))\, (E[q \mid r, a, s = 0] - E[q \mid r, b, s = 0])\,.$$

Interpretation: A Bayesian reacts only if the message changes the inferred probability of the private signal and if the private signal is predictive of quality in the target situation. Under the uninformative persuasion assumption, Bayesians do not react.

3.3 Coarse categories

The coarse thinker has two possible categories:

$$C_1 = \{0, 1\}, \qquad C_2 = \{0, 2\}.$$

The target situation $s = 0$ can be co-categorized with situation $s = 1$ or situation $s = 2$, but not both at once. The category chosen depends on public information and message:

$$C(r, m) = \arg \max_{C \in \{C_1, C_2\}} \widehat{p}(s \in C \mid r, m).$$

Interpretation: The message can matter even when it is uninformative because it changes which analogy or mental model is activated. "Be bullish" can make investing look like grabbing an opportunity; "a tradition of trust" can make investing look like hiring professional advice.

3.4 Coarse expectation of quality

If $C_i = C(r, m)$, the coarse thinker's expectation is

$$E_{C_i}[q \mid r, m, s = 0] = E[q \mid r, m, s = 0]p(s = 0 \mid C_i) + E[q \mid r, m, s = i]p(s = i \mid C_i).$$

Interpretation: The coarse thinker partially imports inference from the co-categorized situation. This is the mathematical expression of associative thinking.

3.5 Persuader's strategy and costs

The persuader observes private signal x and can either report it honestly, $m = x$, at zero cost, or replace it with $m \neq x$ at cost $c \geq 0$. The payoff to the persuader is the audience's expected quality assessment minus any cost of message creation.

Interpretation: The persuader is rational and strategic. Irrationality is in the audience's coarse categorization, not in the persuader's behavior.

4 Models and theoretical specifications

4.1 Uninformative persuasion assumption

The private signal is uninformative about quality in the target situation:

$$E[q \mid r, x, s = 0] = E[q \mid r, s = 0] \quad \text{for all } r, x.$$

Interpretation: This assumption isolates the behavioral channel. If messages still persuade, they do so through coarse thinking rather than objective information.

4.2 Lemma: sophisticated and face-value coarse thinkers

The paper proves that, under the assumptions, the category rule is independent of the persuader's strategy, and sophisticated and face-value audiences hold the same expectations about quality conditional on messages. Therefore, the persuader's equilibrium strategy can be studied as if the audience takes messages at face value.

Interpretation: This lemma simplifies the model. The crucial effect is not whether the audience is naively trusting the persuader, but whether the audience applies a coarse category model once a message is observed.

4.3 Proposition 1: Bayesians are not persuaded by uninformative messages

When all messages are uninformative in situation $s = 0$, a Bayesian audience does not change its assessment in response to the persuader's message. Therefore the persuader never pays a positive cost to fabricate messages for Bayesians.

Economic message: This is the baseline contrast. If persuasion works in the model, it is because the audience is coarse, not because the persuader has conveyed relevant information.

4.4 Proposition 2: Transference

Suppose categorization is fixed and $m = a$ is more favorable than $m = b$ in the co-categorized situation $s = 1$. Then, if

$$c < (E[q \mid r, a, s = 1] - E[q \mid r, b, s = 1])\, p(s = 1 \mid C_1) \equiv c^*,$$

the persuader pays to replace $x = b$ with $m = a$.

Interpretation: Even if the message has no informational value in the target situation, it can be valuable because the audience transfers its meaning from a related situation.

Economic message: This formalizes why putting "silk" in shampoo can work: silk is valuable in the co-categorized situation of hair, not in the situation of shampoo quality.

4.5 Corollary 1: low-involvement products are easier to persuade

The cost threshold c^* rises when the co-categorized situation receives more weight or when the message reaction in that situation is larger.

Interpretation: Low-involvement products occupy a small part of the relevant mental category, so the audience imports more from other situations. This creates greater scope for persuasive but uninformative messages.

4.6 Proposition 3: Framing

Suppose message $m = a$ is pivotal and changes the category from C_1 to C_2. If

$$c < E[q \mid r, a, s = 2]p(s = 2 \mid C_2) - E[q \mid r, b, s = 1]p(s = 1 \mid C_1),$$

then the persuader replaces $x = b$ with $m = a$.

Interpretation: Framing is stronger than transference. The message need not be informative even within a category; it can matter because it changes which category is used.

Economic message: "We try harder" can make Avis's second-place status look like underdog effort instead of market failure.

4.7 Corollary 2: uninformative messages can be optimal

Even if every message is uninformative about quality within every category, a persuader may still pay to send a message that induces a favorable category.

Interpretation: This result captures the pure case of framing. The message changes the interpretive lens, not the facts.

4.8 Corollary 3: withholding good messages

A persuader may withhold or replace a favorable message if the favorable message activates an unfavorable category.

Interpretation: This is the subtle prediction used in the mutual fund evidence. A fund may avoid reporting good relative returns in bad markets because talking about returns makes investors think of stock investing as grabbing an opportunity, a bad frame in down markets.

4.9 Product branding

The paper applies the model to a monopolist selling two homogeneous wines. If coarse thinkers co-categorize U.S. and French wine, the producer may label one wine "Burgundy" and sell it at a higher price even though the wine is objectively the same.

Interpretation: Branding emerges as paid creation of a category association. A brand can be valuable because it imports meaning from another category.

5 Identification and empirical design

5.1 Theory-to-evidence logic

The empirical section does not estimate a structural model. It tests a distinctive qualitative prediction: mutual funds should report past returns when aggregate returns are high, but should avoid reporting even favorable relative returns when aggregate returns are low.

5.2 Advertising data

- The authors collect financial advertisements from *Business Week* from 1994 to 2003 and *Money* from 1995 to 2003.
- The sample includes 1,469 ads from *Business Week* and 4,971 ads from *Money*.
- Business-to-business advertisements are excluded to focus on persuasion of investors.
- The key variable is whether stock mutual fund advertisements include information about the fund's own past returns.

Interpretation: The empirical design studies advertisement content, not just advertising volume. This matches the model, which is fundamentally about message content.

5.3 Market-return conditioning

The key predictor is lagged rolling quarterly S&P 500 return. The model predicts that in high-return periods, funds frame investing as grabbing opportunities and use past return data; in low-return periods, funds avoid this frame and do not use returns even when relative performance is favorable.

5.4 T. Rowe Price test

The T. Rowe Price analysis tests whether the disappearance of past-return advertising after 1999 is merely lack of good news. The authors show that T. Rowe Price had many stock funds outperforming the S&P 500 after 1999, but still largely stopped advertising returns.

Interpretation: This is the key test of Corollary 3. It distinguishes coarse-thinking framing from a simpler "advertise good news" story.

6 Section-by-section study guide

- **Introduction.** Establishes the contrast between economic models of informative persuasion and psychological/marketing views of metaphor, analogy, and framing.
- **Model.** Defines Bayesian and coarse thinkers. The key formal objects are categories C_1, C_2, the category rule $C(r, m)$, and coarse expectations $E_C[q \mid r, m, s = 0]$.
- **Persuasion.** Derives results for Bayesians, transference, framing, and withholding good messages.
- **Product branding.** Shows how uninformative labels can create differentiation and price premia.
- **Mutual funds.** Applies the model to fund advertising and tests the prediction that past-return disclosures are procyclical.
- **Conclusion.** Emphasizes that persuasion can work by changing categories and analogies, not just by changing information.

7 Figures and theoretical result blocks

7.1 Figure I: Timeline of Decision Problem in $s = 0$

Read:

- The individual first observes public signal $r \in \{u, d\}$.
- The individual then observes the persuader's message $m \in \{a, b\}$.
- The individual finally makes a decision based on the assessment of quality q.

Detailed interpretation: The timeline clarifies the order of information and persuasion. Public information arrives before the persuader's message, so framing can change how the public information is interpreted. The audience never sees the original private signal x if the persuader replaces it with m.

Economic message: The model is about how messages reshape inference, not merely about adding facts.

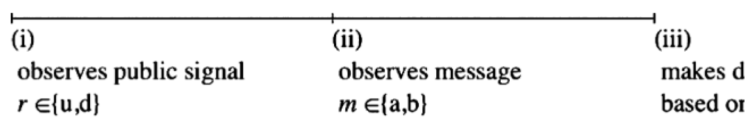


FIGURE I
Timeline of Decision Problem in $s = 0$

Figure 1: Figure I: Timeline of Decision Problem in $s = 0$. The persuader's message enters between public information and the final decision.

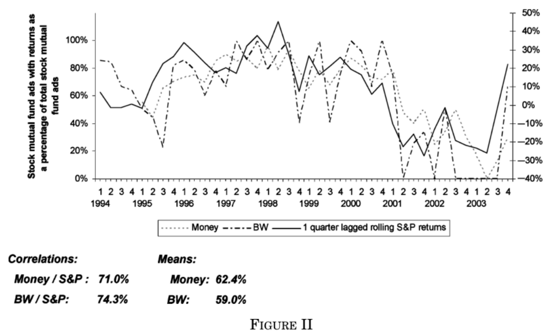
7.2 Figures II and III: Mutual fund advertising evidence

Read:

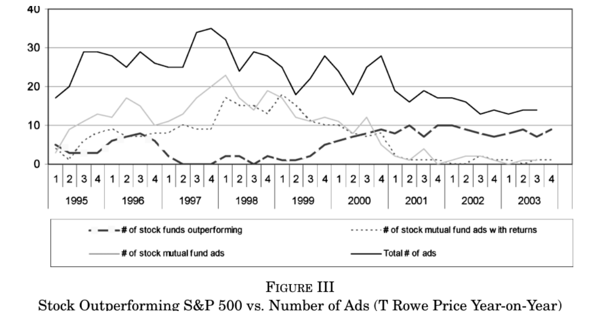
- Figure II plots the share of stock mutual fund ads that report own past returns in *Money* and *Business Week*, together with lagged rolling S&P 500 returns.
- Figure III plots T. Rowe Price stock mutual fund advertising and the number of T. Rowe Price funds outperforming the S&P 500.
- In Figure II, the share of ads using past-return data is strongly procyclical and collapses after the market downturn.
- In Figure III, T. Rowe Price has many funds with good relative performance after 1999, but still reduces return-based advertising.

Detailed interpretation: Figure II supports the basic prediction that funds report returns when aggregate returns are favorable. Figure III is more diagnostic: it shows that the decline in return advertising cannot simply be explained by lack of good relative-return news. The fund complex has good relative-performance facts available, but avoids presenting them because doing so may prime the unfavorable "grabbing opportunities" frame in a down market.

Economic message: The evidence supports the model's distinctive prediction that firms may withhold good information when the information activates an unfavorable category.



(a) Figure II. Stock Mutual Fund Ads Returns / Total Stock Mutual Funds Ads



(b) Figure III. Stock Outperforming S&P 500 vs. Number of Ads

7.3 Theoretical result block 1: Bayesian benchmark

- Bayesians react to messages only if messages change beliefs about the signal and the signal predicts quality in the target situation.
- Under the uninformative persuasion assumption, Bayesians do not react to persuader-created messages.
- Therefore, costly uninformative persuasion is not valuable for Bayesian audiences.

Economic message: The Bayesian benchmark makes the behavioral mechanism transparent: the entire effect comes from coarse categorization.

7.4 Theoretical result block 2: Transference

- Transference occurs when a message that is informative in a co-categorized situation is treated as informative in the target situation.
- The benefit of transference is larger when the co-categorized situation receives more category weight or when the message is more diagnostic there.
- This explains irrelevant product attributes such as silk in shampoo or emotionally charged rhetoric in politics.

Economic message: Persuasion can work by importing meaning from a nearby category.

7.5 Theoretical result block 3: Framing

- Framing occurs when the message changes the category used to interpret the object.
- A message can affect beliefs even if it is uninformative about quality in every situation.
- Successful framing changes the sign or intensity of how public information is interpreted.

Economic message: The persuader does not need to prove quality; he needs to choose the comparison class.

7.6 Theoretical result block 4: Withholding favorable information

- A good message may be withheld if it activates a bad frame.

- In mutual funds, past returns are useful in framing investing as grabbing an opportunity, but that frame is unattractive after aggregate market declines.
- This predicts that funds may avoid reporting returns even when relative performance is good.

Economic message: Message omission can be rational persuasion, not just hiding bad news.

8 Detailed result map

- **Model setup:** individuals evaluate an object after public information and a persuader-controlled message.
- **Bayesian benchmark:** uninformative messages do not change beliefs.
- **Coarse thinking:** individuals use category-level inference, grouping the target situation with one related situation.
- **Transference:** messages can transfer meaning from co-categorized situations.
- **Framing:** messages can change which category interprets the situation.
- **Branding:** uninformative labels can create product differentiation and price premia.
- **Mutual fund evidence:** funds report past returns in bull markets and avoid reporting them in bear markets, even when relative performance is favorable.

9 Short summary

- The paper develops a model of persuasion based on coarse thinking.
- Coarse thinkers apply the same inferential model to all situations in a category.
- Persuaders exploit transference and framing.
- Transference means that meaning from one situation is imported into another.
- Framing means that the message changes the category through which information is interpreted.
- Bayesians do not react to uninformative messages, but coarse thinkers can.
- Product branding can arise from uninformative labels that activate favorable categories.
- Mutual fund advertisers report past returns after good market performance and withhold returns after market declines.
- T. Rowe Price evidence shows that return omission is not just lack of good news.
- The paper explains why persuasive content matters even when it conveys no objective information.

10 Conclusion

10.1 Core conclusion

Mullainathan, Schwartzstein, and Shleifer show that persuasion can work through category-based inference. Audiences may be Bayesian inside a category but choose the wrong category or use one category's inference rule in another situation. This creates room for rational persuaders to use messages that are objectively uninformative.

10.2 Economic intuition

A message is persuasive when it activates a favorable analogy. “Silk” makes shampoo feel like silky hair; “we try harder” makes second place look like underdog effort; return data make mutual fund investing feel like grabbing an opportunity. The message changes what the audience thinks the situation is like.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to formalize a psychological idea - associative/coarse thinking - in an economic model of persuasion. It shows that noninformative persuasion can be optimal, derives comparative statics for when it works, and applies the theory to branding and mutual fund advertising.

10.4 Implications for economics and marketing

The paper suggests that advertising content matters because content selects categories. Firms should not only ask what information to disclose; they should ask what mental representation the disclosure activates. Regulators and economists should recognize that truthful but irrelevant messages can still distort consumer beliefs.

10.5 Limitations

The empirical evidence is illustrative rather than a full structural test. The model abstracts from heterogeneous consumers, repeated persuasion, competition among persuaders, and endogenous learning about categories. Still, the theory provides a disciplined framework for studying persuasive content.

10.6 Takeaway

The main takeaway is that persuasion works by shaping analogies. People do not only update beliefs from facts; they decide, often coarsely, what situation they are in. Persuaders exploit that categorization step.